



MCS3301 – Control Engineering I

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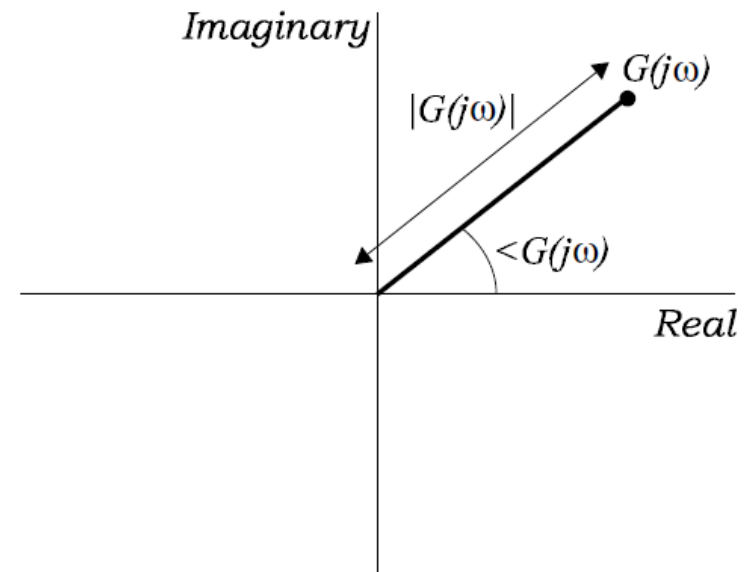


Frequency Response Analysis

- The frequency response examines the output response when the input $u(t)$ is a **sine** wave.
- If the input to a system is sinusoidal, then provided the system is stable, the output is also sinusoidal with a change in **amplitude** and **phase** shift.
- That is if $u(t) = \sin(\omega t)$ then provided $G(s)$ is **stable** and ignoring initial transient conditions

$$y(t) = |G(j\omega)| \sin(\omega t + \angle G(j\omega))$$

- $G(j\omega)$ is called the **gain** of G and $\angle G(j\omega)$ is the **phase** of G at the frequency ω .





First Order Systems

- Suppose that $G(s) = \frac{k}{s+a}$

- Then $|G(j\omega)| = \left| \frac{k}{s+a} \right|$

$$\begin{aligned} &= k \left| \frac{1}{a+j\omega} \times \frac{a-j\omega}{a-j\omega} \right| \\ &= k \left| \frac{a}{a^2 + \omega^2} - j \frac{\omega}{a^2 + \omega^2} \right| \\ &= k \sqrt{\left(\frac{a}{a^2 + \omega^2} \right)^2 + \left(\frac{\omega}{a^2 + \omega^2} \right)^2} \\ &= k \sqrt{\frac{a^2 + \omega^2}{(a^2 + \omega^2)^2}} \\ &= \frac{k}{\sqrt{a^2 + \omega^2}} \end{aligned}$$

$$\begin{aligned} \angle G(j\omega) &= \tan^{-1} \left[\left(-\frac{\omega}{a^2 + \omega^2} \right) / \left(\frac{a}{a^2 + \omega^2} \right) \right] \\ &= \tan^{-1} [-\omega/a] \\ &= -\tan^{-1} [\omega/a] \end{aligned}$$



First Order Systems

- The figures below show the output responses of a first order system $G(s) = \frac{1}{s+1}$ to sine inputs at frequencies of $\omega = 0.1, \omega = 1$ and $\omega = 10$.

- It can be observed that as the input frequency increases, amplitude attenuation and phase shift increase.

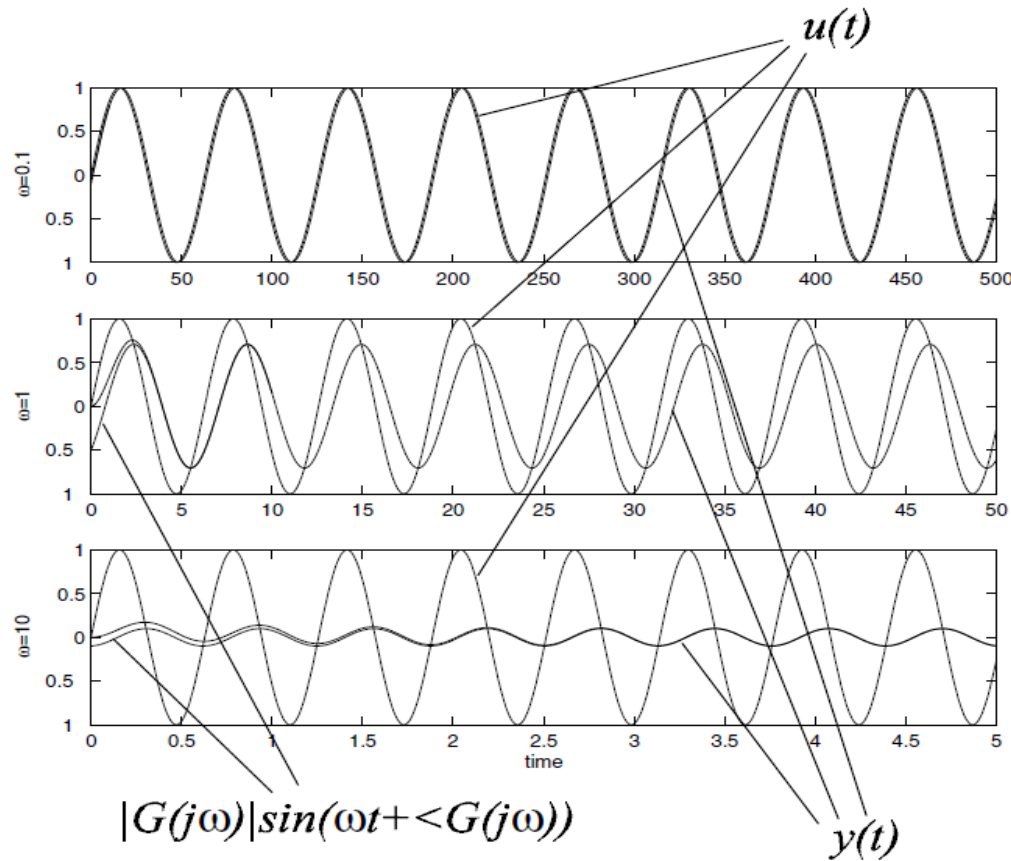
- $y(t) = |G(j\omega)| \sin(\omega t + \angle G(j\omega))$ after the transient phase.

- When ω is small

$$|G(j\omega)| \approx \frac{k}{a} \text{ and } \angle G(j\omega) \approx 0^\circ$$

- When ω is large

$$|G(j\omega)| \approx \frac{k}{\omega} \text{ and } \angle G(j\omega) \approx -90^\circ$$





Bode Plot

- The Bode plot shows gain and phase against frequency on separate graphs.
- Suppose

$$\begin{aligned} G(j\omega) &= G_1(j\omega) \times \frac{G_2(j\omega)}{G_3(j\omega)} \\ &= (g_1 e^{j\phi_1}) \times (g_2 e^{j\phi_2}) \div (g_3 e^{j\phi_3}) \\ &= \frac{g_1 g_2}{g_3} e^{j(\phi_1 + \phi_2 + \phi_3)} \end{aligned}$$

- Then the gain is given by

$$|G(j\omega)| = \frac{g_1 g_2}{g_3}$$

- Also,

$$\log|G(j\omega)| = \log|g_1| + \log|g_2| - \log|g_3|$$

- As such, it is useful to show the gain on a logarithmic graph.
- Similarly the phase of G is given by $\angle G(j\omega) = \phi_1 + \phi_2 + \phi_3$



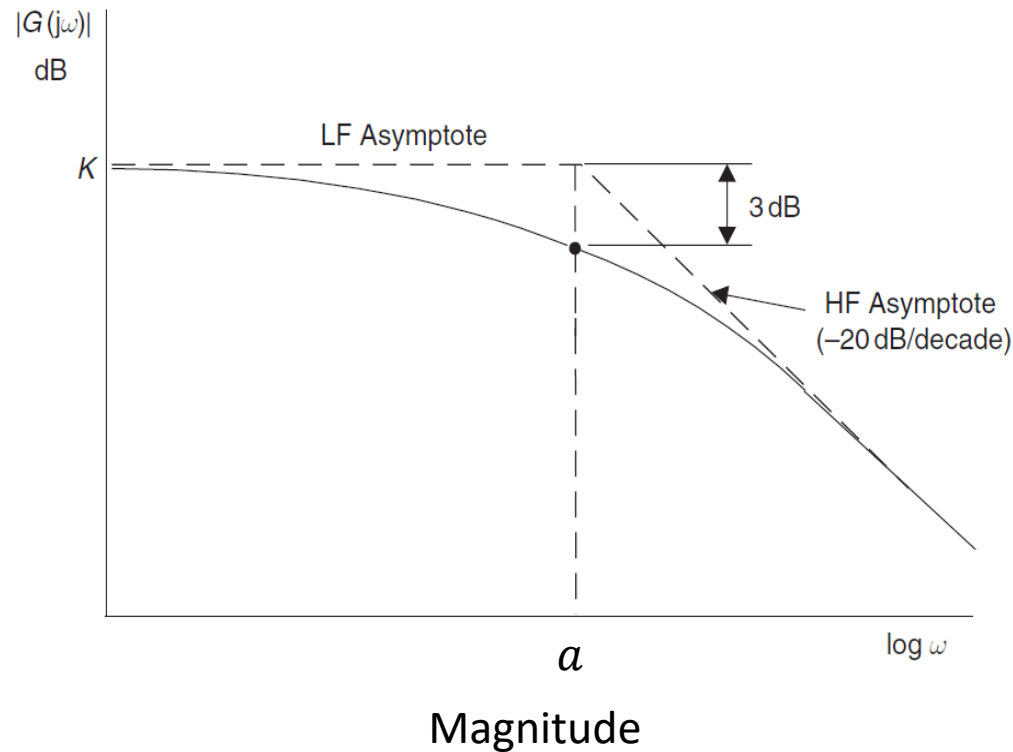
Bode Plot

- The **gain** is usually shown in **decibels** defined as $|G|_{dB} = 20 \log_{10}|G|$
- For a first order system $G(s) = \frac{k}{s+a}$ when ω is small,
$$|G(j\omega)|_{dB} \approx 20 \log_{10} \left(\frac{k}{a} \right)$$
- That is, a steady state **gain** of 1 corresponds to 0_{dB} at **low** frequency.
- When ω is large $|G(j\omega)|_{dB} \approx 20 \log_{10}(k/\omega)$. For instance if $\omega = 10a$ then $|G(j10a)|_{dB} \approx 20 \log_{10}(k/a) - 20$ and if $\omega = 10^2 a$ then $|G(j10^2 a)|_{dB} \approx 20 \log_{10}(k/a) - 40$
- On a log-log scale, the **gain** follows a **horizontal line** $20 \log_{10}(k/a)$ and the drops at **20 dB** per **decade** for **large** frequencies.
- It deviates from these two asymptotes where they meet at $\omega = a$ known as the **break frequency**.
- At this point $|G(j\omega)|_{dB} = 20 \log_{10} \left(\frac{k}{\sqrt{2}a} \right) = 20 \log_{10}(k/a) - 20 \log_{10}\sqrt{2}$
$$\approx 20 \log_{10} \left(\frac{k}{a} \right) - 3 \text{ dB}$$



Bode Plot: Magnitude

- The **magnitude** plot consists of a **low frequency** asymptote and a **high frequency** asymptote



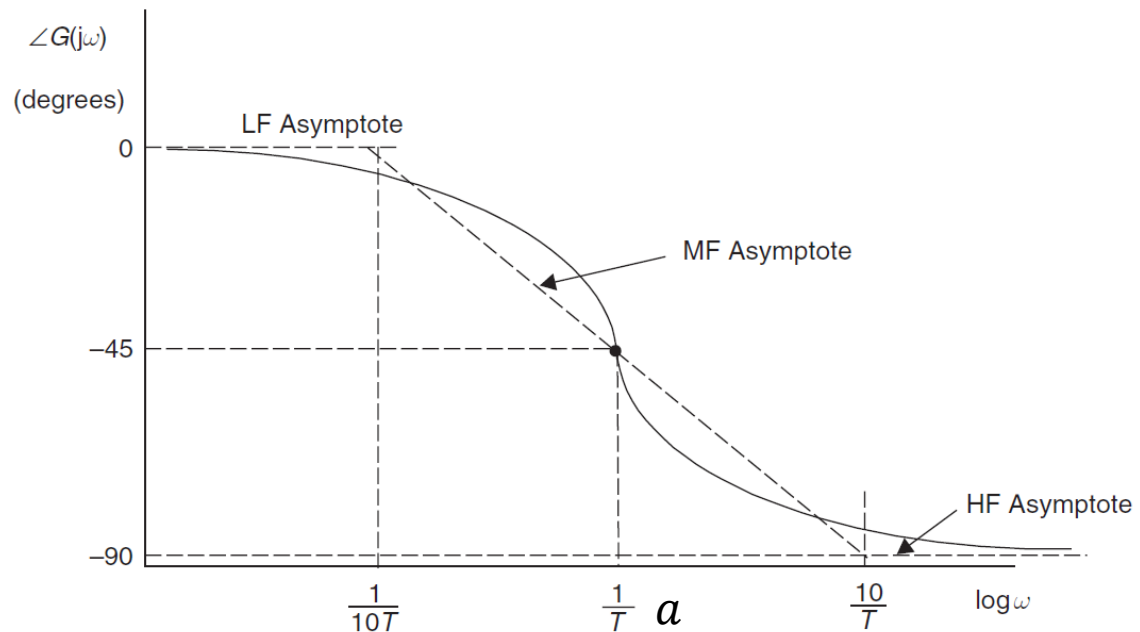


Bode Plot: Phase

- The **phase** on a Bode plot is drawn on a **linear** scale against **frequency** on a **logarithmic** scale.
- It has a **low** frequency asymptote at 0° a **high** frequency asymptote at -90° .
- It has a mid frequency asymptote that intersects the LF asymptote at $a/10$ and the HF asymptote at $10a$
- At $\omega = a$, $\angle G(ja) = -\tan^{-1}(a/a) = -45^\circ$

Phase plot

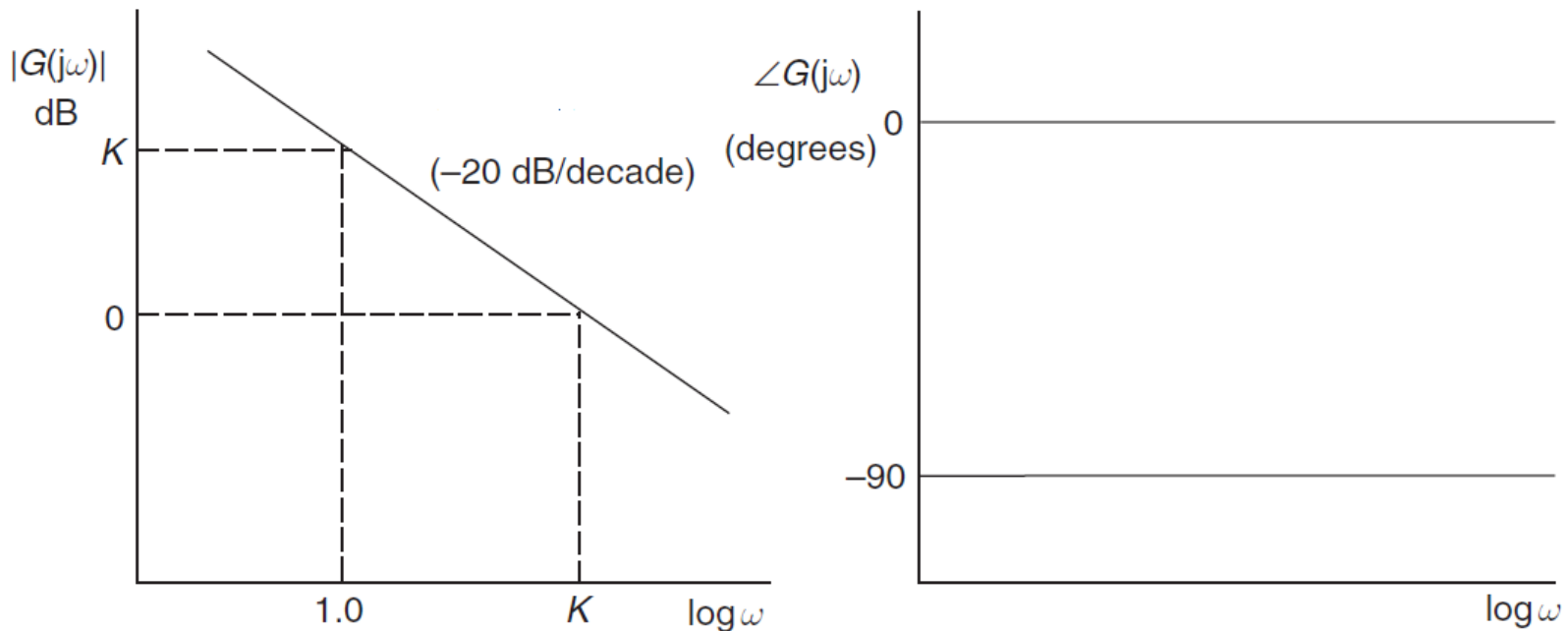
$$\frac{a}{10}$$





Bode Plot: Example 1

- Consider a pure integrator of the form $G(s) = \frac{k}{s}$
$$G(j\omega) = \frac{k}{j\omega}, |G(j\omega)| = \frac{k}{\omega}$$
- When $\omega = 1$, $|G(j\omega)| = k$ dB
- When $\omega = k$, $|G(j\omega)| = 1 = 0$ dB
- $\angle G(j\omega) = -\tan^{-1}(\omega/0) = -\tan^{-1} \infty = -90^\circ$

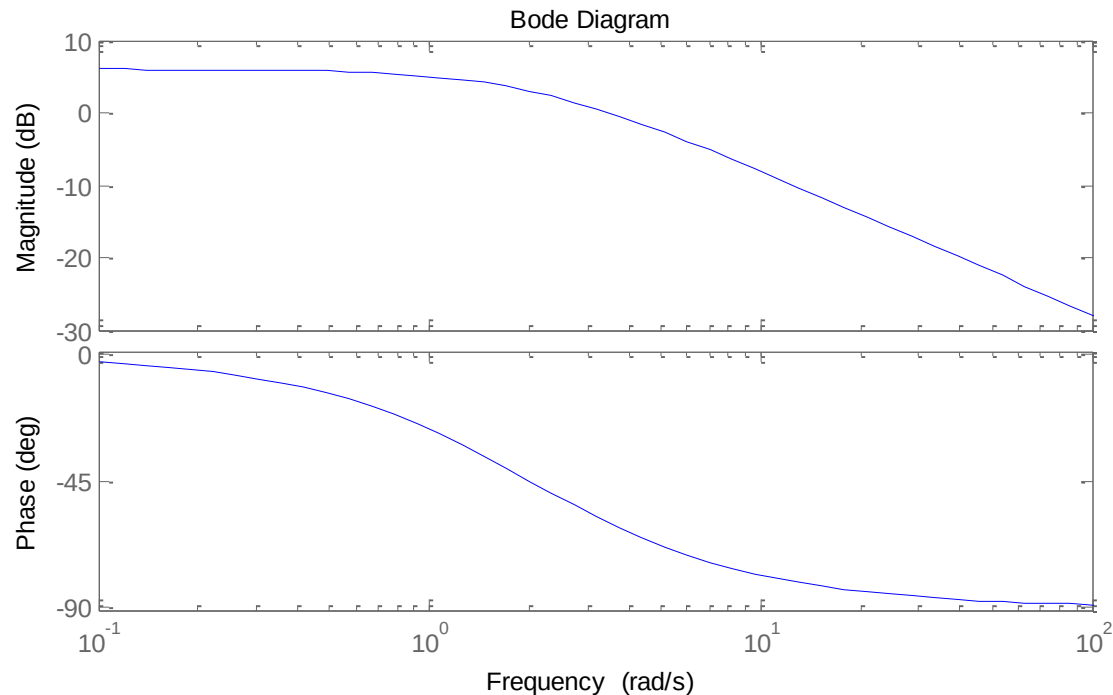


Bode diagram of a pure integrator $G(s) = \frac{k}{s}$



Bode Plot: Example 2

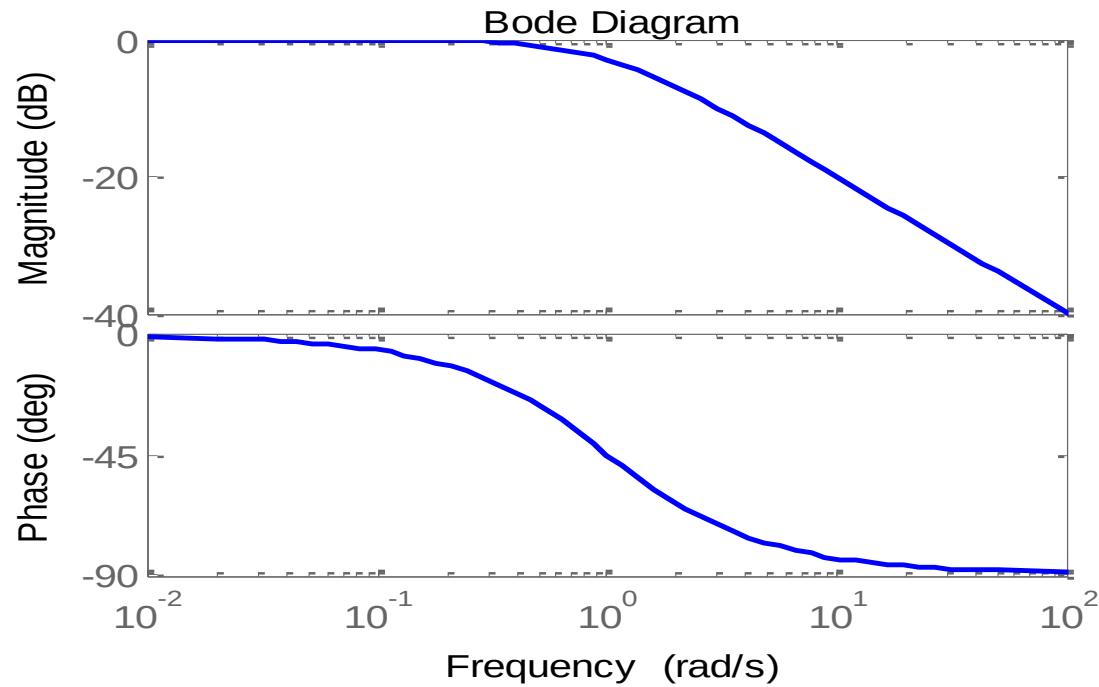
- Construct a bode diagram for the system $G(s) = \frac{2}{1+0.5s}$
- The low frequency asymptote is at 6 dB
- Asymptote intersection occurs at 2 rads/sec
- $\angle G(j\omega) = -\tan^{-1}$





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Bode Plot

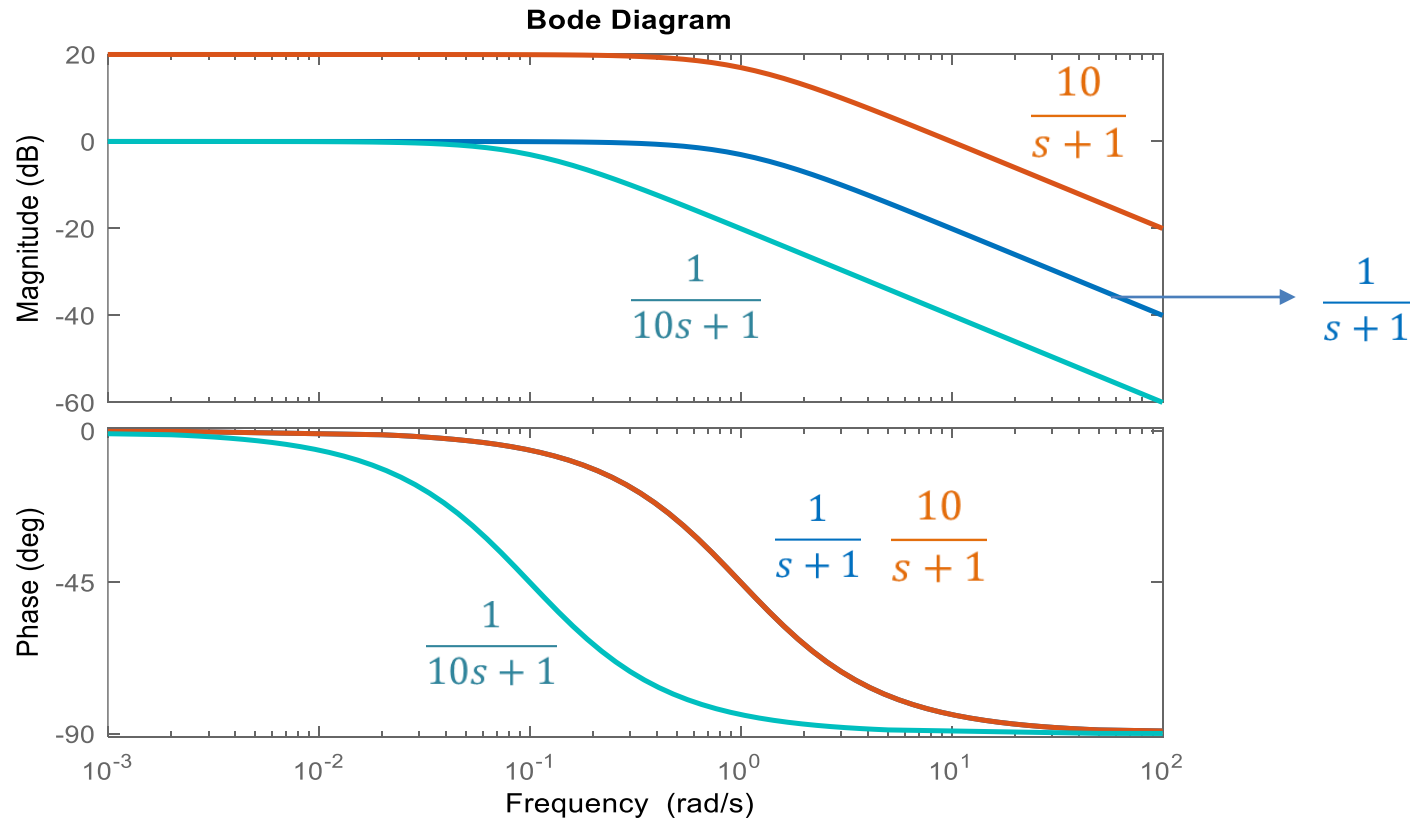


Bode diagram of a first order plant $G(s) = \frac{1}{s+1}$



Bode Plot

- The bode diagram of $\frac{k}{s+1}$ is the same with $\frac{1}{s+1}$ but moved up by k .



Bode diagram of a first order plant $G(s) = \frac{1}{s+1}$, $G(s) = \frac{10}{s+1}$, $G(s) = \frac{1}{10s+1}$



Second Order Systems

- A Consider the second order transfer function

$$G(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Its frequency response at ω is

$$\begin{aligned} G(j\omega) &= \frac{k\omega_n^2}{\omega_n^2 - \omega^2 + 2j\zeta\omega_n\omega} \\ &= \frac{k}{1 - \omega^2/\omega_n^2 + 2j\zeta\omega/\omega_n} \end{aligned}$$

- When ω is small

$$\begin{aligned} G(j\omega) &\approx k \\ |G(j\omega)| &\approx 20 \log_{10} k \text{ dB} \\ \angle G(j\omega) &\approx 0^\circ \end{aligned}$$



Second Order Systems

- When ω is large

$$G(j\omega) \approx -\frac{k\omega_n^2}{\omega^2}$$

$$|G(j\omega)| \approx (20 \log_{10} k + 40 \log_{10} \omega_n - 40 \log_{10} \omega) dB$$

$$\angle G(j\omega) \approx -180^\circ$$

- When $\omega = \omega_n$

$$G(j\omega) \approx -\frac{k}{2j\zeta}$$

$$|G(j\omega)| \approx (20 \log_{10} k - 20 \log_{10} 2\zeta) dB$$

$$\angle G(j\omega) \approx -90^\circ$$



Second Order Systems: Example

- Construct a Bode diagram for the second order system

$$G(s) = \frac{4}{0.25s^2 + 0.2s + 1}$$

- The transfer function can be expressed as

$$G(s) = \frac{4 \times k}{s^2 + 0.8s + 4}$$

- It is easy to see that $k = 4$, $\omega_n = 2$ and $\zeta = 0.2$
- When ω is small,

$$\begin{aligned} G(j\omega) &\approx k \\ |G(j\omega)| &\approx 20 \log_{10}(4) = 12 \text{ dB} \\ \angle G(j\omega) &\approx 0^\circ \end{aligned}$$

When $\omega = \omega_n$

$$\begin{aligned} G(j\omega) &\approx -\frac{k}{2j\zeta} \\ |G(j\omega)| &\approx (20 \log_{10}(4) - 20 \log_{10} 2(0.2)) = 20 \text{ dB} \\ \angle G(j\omega) &\approx -90^\circ \end{aligned}$$



Second Order Systems: Example

- When ω is large i.e. 100 rad/s

$$G(j\omega) \approx -\frac{k\omega_n^2}{\omega^2}$$

$$|G(j\omega)| \approx (20 \log_{10}(4) + 40 \log_{10}(2) - 40 \log_{10}(100))dB$$

$$|G(j\omega)| \approx 55^0$$

$$\angle G(j\omega) \approx -180^0$$

